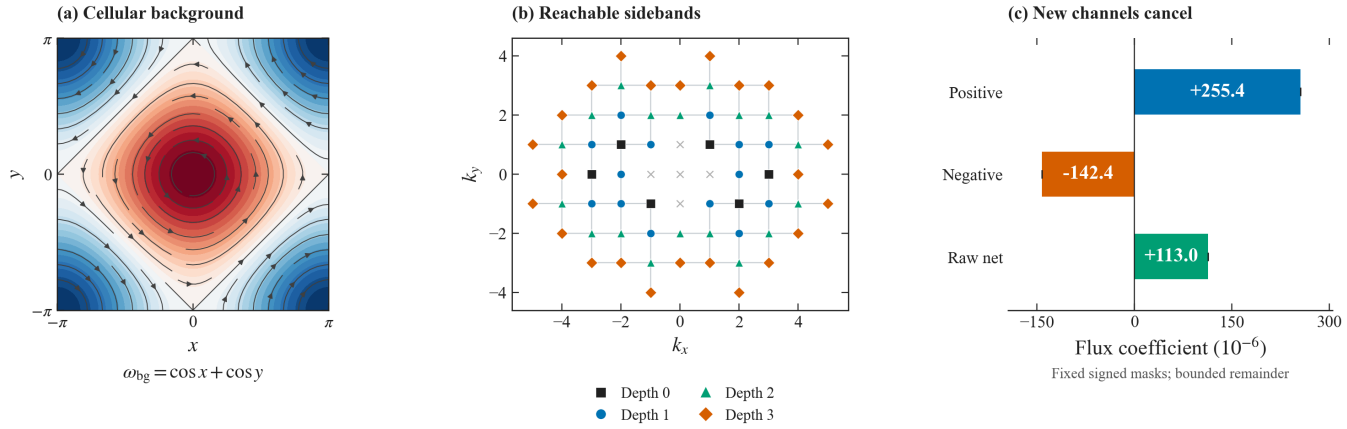
How a coherent vortex reshapes triad energy transfer

An order-one cellular vortex reorganizes connected three-point correlations. A controlled two-dimensional Navier-Stokes model quantifies the resulting flux shift with analytic proofs and exact-arithmetic bounds.

10.77%
attenuation of negative flux

>99.43%
of the shift is connected

62 modes
three sideband layers



Analytic background (not DNS), certified sideband depths 0-3, and new signed-channel centers in units 10^{-6}. Exact error bounds accompany the source data.

1 / Why isolated-triad intuition can fail

Waleffe links isolated-triad stability to turbulent transfer [1]. Recent phase closures replace neighboring triads with weakly correlated noise [2, 3]. Strong coherent vortices challenge this approximation by strengthening modal correlations [3].

Card A isolates feedback. In a fixed-amplitude four-shell Sabra phase model, coupled phases acquire a positive bias at a neutral point where the isolated prediction is zero; nearby, transfer signs disagree. Card B moves to a spatial PDE with evolving amplitudes.

2 / A maintained cellular vortex plus fluctuations

$$\omega = A(\cos x + \cos y) + \varepsilon\zeta, \qquad \hat{\zeta}_k = 0 \quad (|k| \le 1)$$

On the periodic torus, A need not be small; epsilon controls fluctuation nonlinearity. With G the inverse negative Laplacian, J(f,g) = $f_y g_x$ - $f_x g_y$, and P_K retaining 1 < |k| <= K, the normalized outward flux is

$$T_K(\zeta) = \langle GP_K\zeta, J(G\zeta,\zeta)\rangle, \qquad \mathcal{F}_{K,N}(\varepsilon) = \varepsilon\mu_{\varepsilon,N}(T_K)$$

Only three fluctuation legs enter this observable. Viscous smoothing $H^{-3/2}$ to L^2 has integrable short-time singularity $t^{-3/4}$. Together with uniform energy moments and Poisson reduction, it gives

$$|\mathcal{F}_{K,N}(\varepsilon) - \varepsilon^2 F_{A,K,N}| \le C\varepsilon^4, \qquad F_{A,K,N} \longrightarrow F_{A,K}$$

C is independent of the Galerkin cutoff N for fixed parameters. The certificates concern the leading coefficient F; finite-epsilon accuracy also includes the remainder.

Where the 10.77% shift comes from

Set viscosity and background amplitude to one, with zero linear drag and cutoff K = 2. Force the four modes ±p, ±q with p = (3, 0), q = (-2, 1), and covariance weights 9 and 5. Their interaction feeds the low mode p + q = (1, 1). With no background, $F_{0,2} = -1/225$ exactly.

Infinite-model coefficient	Certified outward enclosure
$F_{1,2}$	[-0.003965854562651, -0.003965845004717]
Delta $F_1 = F_{1,2} + 1/225$	[0.000478589881793, 0.000478599439728]

F remains negative; its change is strictly positive. The negative-flux amplitude decreases by 10.768272%-10.768488%. This is attenuation, without a direction reversal. The absolute coefficient error radius is below $\mathbf{4.78 \times 10^{-9}}$.

3 / Connected response supplies almost all of the change

$$F_A = t\kappa_{3,A} + t\,\mathrm{Sym}(m_{1,A} \otimes U_A) = F_A^{\mathrm{conn}} + F_A^{\mathrm{mean}}$$

U_A is the Gaussian reference covariance; $m_{1,A}$ is the first nonlinear mean response. Sym is the sum of three placements. The connected term removes the part of the third moment explained by mean times covariance. It retains amplitude-phase organization that cannot be reduced to a mean-flow artifact.

The mean contribution begins at order A^4. The leading A^2 correction is therefore entirely connected. At A = 1, the certified decomposition is:

Contribution to the shift	Center × 10^6	Certified share (%)
Original connected-triad attenuation	+362.477	75.736047-75.739564
Other connected triads, net	+113.419	23.697093-23.699571
Induced mean response	+2.699	0.563853-0.563872

The original triad supplies about 75.7% of the shift. The other channels supply about 23.7% net, but their raw positive and negative groups are much larger: **+255.4** and **-142.4**, in units 10^{-6}. Their compensation leaves 113.0; removing the complement's mean term gives the connected net 113.4.

These groups use the fixed B-M01 sign masks. The unlisted complement remains enclosed by an explicit error bound; the construction does not classify every unknown infinite-tail orbit.

How the numerical claim becomes a certificate

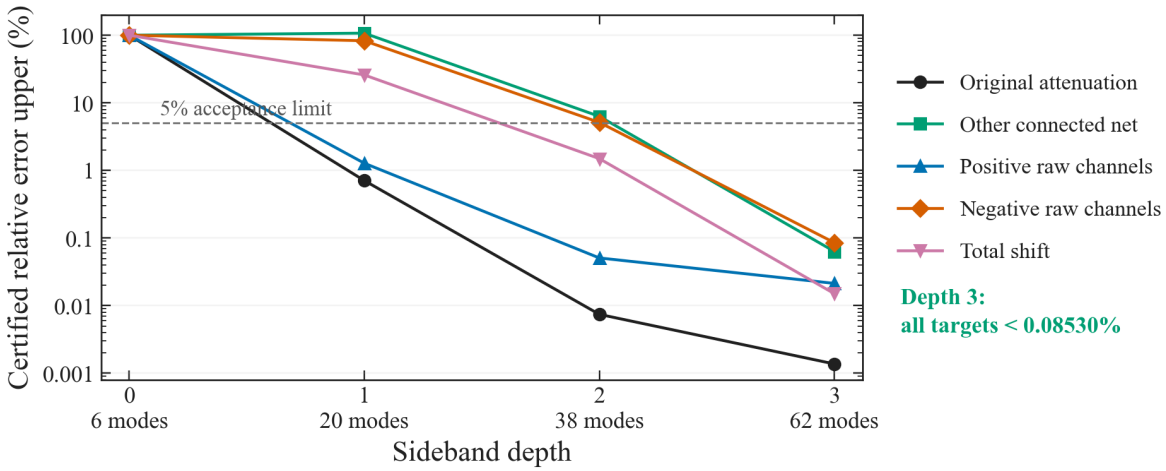
Sparse rational moment candidates are tested on the entire infinite lattice, including every generated exterior output. A sharp dual norm of 1/720 converts residual bounds into scalar flux error. Independent code checks the same result; floating-point convergence alone is not the certificate.

Error budget	Approximate rigorous bound
Finite solve	$6.031742283070 \times 10^{-10}$
Exterior + source pruning	$4.175792455590 \times 10^{-9}$
Directed arithmetic	$1.538205296343 \times 10^{-24}$

Acceptance uses saved exact fractions. The total radius is $4.778966683897 \times 10^{-9}$. The exported checker reconciles those fractions; it does not rerun the full nonlinear residual construction.

Three layers restore the signed balance

The cellular vortex shifts wavevectors by $\pm e_x$ and $\pm e_y$ wherever the coupling is nonzero. Starting from the original triad and its conjugates, B-R01 adds successive coupling-graph neighborhoods. Forcing, equation coefficients and observable stay fixed; no flux coefficients are fitted.



Points are certified relative-error upper bounds against the infinite-model reference. All five targets must meet the 5% rule. Separate lower bounds prove failure of depths 0-2.

The 6-mode model predicts no shift. At 20 modes, original attenuation is within 0.71%, yet negative-channel error exceeds 82.73%. At 38 modes, it still exceeds 5.1264%. Only 62 modes pass all five targets, each below 0.08530%.

62 modes is minimal within this declared family, not over all reductions. Matching the original triad alone can hide a badly distorted signed balance.

4 / What this suggests for LES and reconstruction

Matching E(k) constrains amplitudes, while transfer also depends on connected, amplitude-weighted phase organization. A coherent vortex mixes frequencies: early truncation breaks signed compensation. Check connected response, both signed channels and net flux together.

Internal triad phase is translation-invariant; spatial pairing additionally tests vortex position. A 2D-3C diagnostic bridge uses common in-plane filtering, conditioned on coarse strain/vorticity:

$$\Pi_\ell^{\parallel} = -\sum_{i=1}^{3}\sum_{\alpha=x,y} \tau_{i\alpha}^{\ell}\partial_\alpha \bar{u}_i^{\ell}, \quad \tau_{i\alpha}^{\ell} = \overline{u_i u_\alpha}^{\ell} - \bar{u}_i^{\ell}\bar{u}_\alpha^{\ell}$$

This is only the in-plane contribution. The controlled 2D theorem supplies a mechanism to test in DNS; it is not a universal claim about three-dimensional cascades or all SGS closures.

References & evidence

[1] F. Waleffe (1992). The nature of triad interactions in homogeneous turbulence.
[2] S. J. Benavides & M. D. Bustamante (2026 version). Phase dynamics and their role determining energy flux in hydrodynamic shell models.
[3] S. J. Benavides & M. D. Bustamante (2026 preprint). Triad phase dynamics determine cascade direction in two-dimensional turbulence.

Companion repository: bilingual text, mathematical contract, exact certificates, source hashes and reproducible figures. Underlying research: one Lead, one independent Challenger, 60-second numerical command limit.